PROJECT: DESIGN YOUR AIRPORT

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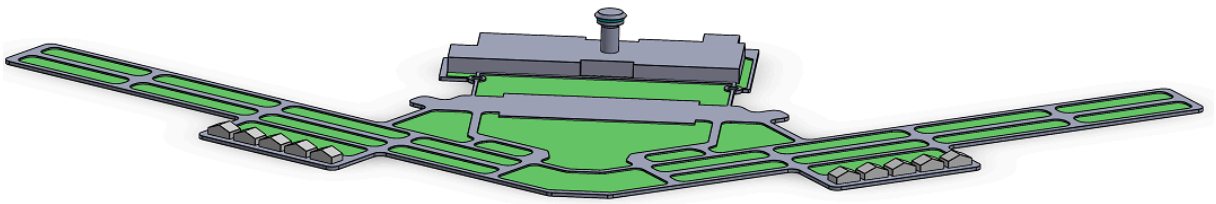
Abstract *The purpose of this project is to implement in a practical situation the knowledge acquired during the class sessions of the course. This paper focuses on determining the essential design parameters and characteristics for the efficient operation of an airport. Key aspects such as runway configuration, taxiway design, terminal infrastructure, apron layout, and landside dimensioning will be thoroughly examined.*



Introduction

In the development of this paper, the following airport characteristics have been considered:

- Reference key: 2C
- Altitude: 100 m
- Reference temperature: 31ºC

When planning an airport, designers must follow a precise set of parameters as outlined in the ICAO document "Aerodrome Design and Operations, Volume I".

Based on the criteria outlined previously, the airport is focused on regional flights, training activities and private jet services. Some similar airports can be Manresa Aerodrome and La Cerdanya Aerodrome.

1. Selection of the reference aircraft [2]

The aerodrome reference code 2C requires, on the one hand, a reference field length of 800 m up to but not including 1200 m. On the other hand, the wingspan ranges from 24 m up to but not including 36 m.

The Bombardier Dash 8 Q300 has been selected as the reference aircraft. It falls under code 2C with a reference field length of 1085 m and a wingspan of 27,43 m.

Figure 1. Bombardier Dash 8 Q300. Source [4]

2. Definition of the Runway Orientation

The ICAO guidelines for runway design state that an aerodrome must maintain a usability factor of at least 95%, meaning that the orientation and number of runways should ensure that aircraft can operate safely with a crosswind component of no more than 19 km/h (10 kt) in normal circumstances. This standard is especially crucial for aeroplanes with a reference field length of less than 1200 m.

Using the wind rose configuration and the excel provided, it can be concluded that the requirement of usability is not possible with a runway by itself. The maximum value of usability possible is 94,5%.

Given this, the aerodrome must consider the trade-offs between increasing usability and containing costs. While the single-runway option might be more economical, it has a reduced usability factor, possibly below what is deemed acceptable for safe and efficient operations. This could lead to higher risks of delays, diversions, or cancellations due to adverse weather conditions.

Implementing two runways to achieve the desired usability factor would significantly escalate the cost, particularly for an aerodrome categorised as 2C. This classification typically indicates a medium-sized aerodrome with moderate traffic levels.

Given these considerations, the recommended solution is a crossed-runway design with an orientation of E for the primary runway and NNE for the secondary [see the Appendix 2 for details]. By opting for a crossed-runway approach, the aerodrome can improve usability and suggests a compromise that may be more feasible in terms of operational flexibility and costs compared to a single-runway alternative, which would yield an even lower usability rate.

Wind Rose (option 2):

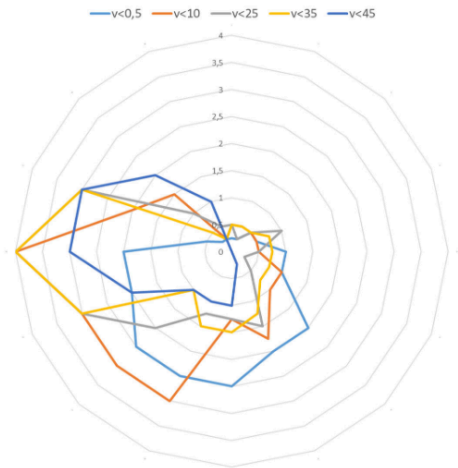


Figure 2. Wind Rose (option 2).

The proposed orientation of the runways is:

- 09-27
- 02-20

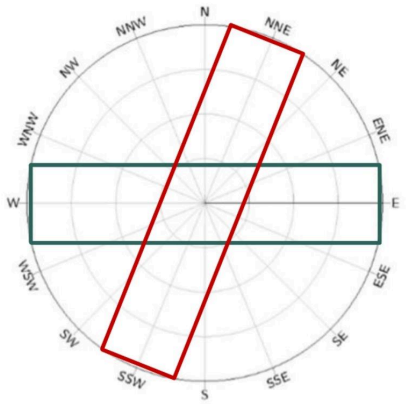


Figure 3.Runway orientation.

To be able to use both runways simultaneously to perform landing and take-off operations, an open V configuration has been chosen. This design increases capacity, reduces congestion, and enhances safety.

The airport is located in Las Terrenas, Dominican Republic, where the magnetic declination is -13º. Because of this, the runway orientations need to be adjusted by 10º. Consequently, the runway designations are 08-26 and 01-19 [see the Appendix 3 for details].

3. Definition of the Airside

In the following sections, the reference aircraft data has been used to perform the required calculations.

3.1. Runway characteristics

The airport runway system consists of two open V configuration runways. Instrumental approaches are allowed due to the fact that the airport has CAT I precision approach runways.

While the CAT I system could theoretically handle aircraft operations every minute, it has been decided to schedule flights with a 1,5-minute interval to allow flexibility for delays or other unforeseen situations [see the Appendix 4 for details].

3.1.1. Length

Having a reference field length of 1085 m at sea level under standard atmospheric conditions, zero wind, and zero slope, correction factors must be applied to determine the actual length of the runway. The airport criteria specified includes an elevation of 100 m and a reference temperature of 31ºC. The corrections are applied as follows:

Altitude correction:

$$L_h = L \cdot (1 + \frac{0{,}07 \cdot h(m)}{300})$$

$$L_h = 1085\, m \cdot (1 + \frac{0{,}07 \cdot 100\, m}{300}) = 1110{,}32\, m$$

Temperature correction:

$$T_{sh} = 15\, ºC - \frac{6{,}5 \cdot h(m)}{1000}$$

$$T_{sh} = 15\, ºC - \frac{6{,}5 \cdot 100\, m}{1000} = 14{,}35\, ºC$$

$$L_T = L_h \cdot (1 + 0{,}01 \cdot (T_{ref} - T_{sh}))$$

$$L_T = 1110{,}32\, m \cdot (1 + 0{,}01 \cdot (31\, ºC - 14{,}35\, ºC)) = 1295{,}19\, m$$

Slope correction:

Since the assumed longitudinal runway slope is 0%, no slope correction factor is necessary.

When determining the minimum distances required for both take-off and landing, it's crucial to consider the correction factors. During take-off calculations, all correction factors are incorporated, resulting in a minimum required runway length of 1295 m. On the contrary, for landing, only the altitude correction is applied, resulting in a minimum runway length of 1110 m.

Since the minimum runway length is determined by the most restrictive value, which corresponds to take-off in this case, the required runway length stands at 1295 m.

3.1.2. Width

According to the Table 5-1 in ICAO Doc. 9157 [3], the runway width specified for category 2C is 30 m.

3.1.3. Longitudinal and transversal slopes

As stated before, the longitudinal runway slope is 0% for simplifications.

Nevertheless, for optimal runway conservation, a transverse slope is necessary. In this scenario, the transverse slope must conform to the highest permissible value specified in ICAO regulations, which is 1,5%.

3.1.4. Strip

To safeguard the runway, a strip must be defined to provide an obstacle-free area (excluding the frangible ones) to ensure safe operations. At a 2C airport, this area should extend 60 m from both thresholds and 70 m from the runway centreline toward the edges.

Moreover, the strip should include a graded section to shield aircraft in case of a centreline deviation. Therefore, the graded portion

of the strip will extend outwards 40 m from the runway centreline.

3.1.5. Shoulders

Since the airport is classified with the reference letter C, there is no requirement to include shoulders at the edges of the runway.

3.1.6. Runway Extreme Safety Area (RESA)

The Runway Extreme Safety Area (RESA) will be designed to reduce the risk of aircraft damage during short landings or runway overruns. It will provide smoother braking and improve accessibility for firefighting and rescue vehicles.

Since the runway supports instrumental approaches, the addition of a RESA is mandatory. In this case, the RESA will extend 120 m from the runway strip limit and will be twice as wide as the runway, measuring 60 m.

3.1.7. Clearway & Stopway

Safety can be enhanced by adding a clearway and a stopway.

A clearway is an obstacle-free area intended to ensure safe takeoff operations, extending beyond the available takeoff run. Its length should not exceed half of the TODA. For this case, it can measure 25% of the TORA, resulting in a clearway of about 325 m. The width of the clearway extends 75 m on either side of the runway's centreline.

A stopway is a rectangular area at the end of the available takeoff distance. It matches the runway's width and is designed to allow an aircraft to stop safely after an aborted takeoff. The stopway's length should not exceed half of the TODA and the stopway is wanted to be shorter than the clearway. To ensure this criteria, a length equal to the 80% of the clearway is taken, yielding an approximate length of 260 m.

3.1.8. Declared distances (TORA, TODA, ASDA, LDA)

Once the length of the runway, the stopway, and the clearway have been determined, the declared distances can be specified. It is important to highlight that both runways will have the same characteristics since they will serve the same operations. Results are rounded to the nearest tenth.

$TORA \simeq 1300\,m$

$TODA = TORA + CWY = 1295 + 325 = 1620\,m$

$ASDA = TORA + SWY = 1295 + 260 \simeq 1560\,m$

$LDA = TORA \simeq 1300\,m$

3.1.9. Runway horizontal markings [5]

Runway markings are crucial for ensuring the safety and efficiency of airport operations.

<u>Runway designation:</u>

The runway designation is: 08-26 and 01-19.

Due to the fact that non-parallel open V configuration runways are considered, the distinction between left and right must not be done. Because instrumental approaches can be performed, the number will be placed above the threshold.

<u>Threshold:</u>

As both runways allow instrumental approaches, a threshold must be added. It consists of a configuration of uniformly sized longitudinal stripes symmetrically arranged with respect to the runway centreline.

The stripes will begin at 6 m from the start of the runway and will have a length of 30 m and a width of 1,8 m. Because both runways have a width of 30 m, there will be a total of eight parallel stripes at the threshold of each runway.

<u>Aiming point and touchdown area:</u>

The aiming point marking is made up of two bands located 250 m from the runway threshold, positioned symmetrically with respect to the centreline. The bands are separated a distance of 18 m between the inner sides. Each band has a length of 30 m and a width of 6 m.

The touchdown area consists of pairs of rectangular signals arranged symmetrically with respect to the runway centreline with a longitudinal separation of 150 m from the threshold.

<u>Runway centreline:</u>

The runway centreline marking guides pilots during landing and takeoff.

It consists of a line of uniformly spaced stripes and gaps, providing visual guidance for accurate alignment with the runway's axis. The centreline begins just after the runway designation markings. The length of the stripes is 30 m, with gaps of 20 m between them. As the runway is CAT I, the stripe width is 0,45 m.

<u>Runway edge:</u>

Two continuous stripes, each 0,9 m wide, run along the edges of the runway to clearly mark the edges.

3.1.10. Lights

Runway lights are of paramount importance when planning an airport because they can improve visibility during night operations and in low-light conditions, such as fog or rain.

Runway edge:

Runway edge lights are installed in two parallel rows along the length of the runway, equidistant from the edge and with intervals of 60 m. These white lights are designed to be clearly visible from all necessary angles, providing guidance to pilots during landings and takeoffs.

Runway centreline:

Runway centreline lights are installed along the runway's axis to guide pilots during takeoff and landing. These lights run from the threshold to the runway's end, with a spacing of about 15 m. The colour pattern of these lights helps indicate runway position: from the threshold to the midpoint, they are white. From the midpoint to 300 m from the end, they alternate between red and white, while the final 300 m are entirely red.

Runway threshold and wing bar:

Runway threshold lights should be placed between the rows of runway edge lights to mark the beginning of the runway. Since precision approach runways require extra visibility, wing bar lights may also be installed to enhance guidance.

Both runway threshold lights and wing bar lights are solid green and are designed to be visible only from one direction, serving as clear markers of the runway's start for incoming aircraft.

Runway end:

Runway end lights are arranged in a line perpendicular to the runway's axis, positioned as close as possible to the runway's end.

There should be a minimum of six lights, spaced between the rows of runway edge lights. These lights are fixed, unidirectional and red, indicating the end of the runway to departing aircraft and ensuring a clear visual signal to pilots.

Touchdown:

The touchdown lights are not necessary for CAT I precision approach runways.

3.1.11. Approach lighting systems

CAT I approach lighting system:

The lighting system extends along the runway's centreline for a distance of 900 m. The first 300 m consist of a single line of lights. In the next 300 m, the lights are arranged in two parallel lines. For the final 300 m, the setup consists of three parallel lines of lights.

Intersecting this centreline, there are five crossbars, spaced every 150 m. These crossbars decrease in length as they get closer to the runway, creating a triangular pattern that points toward the runway threshold.

PAPI:

It is a mechanism which pilots utilise to get guidance when making their final approach to a runway. In order to ensure a safe landing, these lights give pilots visual signals to maintain the proper glide path.

It will consist of a wing bar with four elements located at equal intervals. The system will be placed on the left side of the runway.

3.2. Taxiway characteristics

The taxiway is the connection between the runway and the apron. It is designed to facilitate as much as possible this connection, reducing the time and maintaining the safety levels.

3.2.1. Width

The minimum width of the taxiway is 15 m for aeroplanes with a wheelbase less than 18 m, which applies to the reference aircraft with a wheelbase of 10 m. The minimum width of the taxiway and shoulder is 25 m and the minimum width of the taxiway strip is 52 m. The strip provides an obstacle-free area avoiding any possible damage to the aircraft on taxi.

3.2.2. Longitudinal and transversal slopes

The maximum longitudinal slope of the taxiway is 1,5% with a maximum change in slope of 1% per 25 m. The maximum transversal slope of taxiway pavement is also 1,5%. It is also important to mention that the pavement resistance on the taxiway is at least equal to the one of the served runway.

3.2.3. Separation

Separation distance between apron taxiway/taxiway centreline and apron taxiway/taxiway centreline.

$$V = Y + X + Z$$
$$V = 27,4 + 3 + 5$$
$$V = 35,4\,m$$

Separation distance between apron taxiway/taxiway centreline and object.

$V = ½\,Y + X + Z$
$V = ½ \cdot 27,4 + 3 + 5$
$V = 21,7\,m$

Separation distance between aircraft stand taxilane centreline and aircraft stand taxilane centreline.

$V = Y + gear\ deviation + Z$
$V = 27,4 + 1,5 + 3$
$V = 31,9\,m$

Separation distance between aircraft stand taxilane centreline and object.

$V = ½\,Y + gear\ deviation + Z$
$V = ½ \cdot 27,4 + 1,5 + 3$
$V = 18,2\,m$

Separation distance between runway centreline and taxiway centreline.

$V = ½\,(SW + Y)$
$V = ½ \cdot (140 + 27,4)$
$V = 83,7\,m$

Where,

$V = total\ separation\ distance$
$Y = wingspan$
$X = maximum\ lateral\ deviation$
$Z = increment$
$SW = strip\ width$

Strip width is equal to 140 m because it extends 70 m to each side of the runway centreline.

3.2.4. Exit and entrance

Exit taxiways allow aircraft to move off the runway so another operation can take place.

Exit taxiways are constructed at the end of the runway and at each of its sides in order to connect the taxiway with the apron.

A single runway entrance is sufficient to accommodate the demand for take-offs.

Rapid exit taxiways (RETS):

Their objective is to minimise the runway occupancy time and therefore increase the capacity of the aerodrome. Nevertheless, due to the category of the airport, no rapid exit taxiways are required since the volume of operations is expected to be low.

3.2.5. Curves

Curves should be as few and small as possible and the radius of curve has been set to 30,48 m (from the plane manual) [2]. This radius ensures that when the cockpit of the aeroplane remains over the taxiway centre line markings, the clear distance between the outer main wheels of the aeroplane and the edge of the taxiway is not less than the specified by ICAO, which is 3 m.

3.2.6. Horizontal markings

Centreline and edge marks:

The centreline mark and the taxiway edge mark are yellow coloured and surrounded by a contrast black strip. The width of the centreline mark is 20 cm and the width of the edge line is 15 cm. The contrast strips are half the width of the yellow markings.

Holding points marks:

Consists of a simple yellow dashed line. Designates the point at which an aircraft has to wait before entering a taxiway intersection.

Paved shoulder marks:

It is used to indicate that certain zones looking like pavement are not prepared to support the weight of aircraft. It consists of yellow transverse strips placed in the non-resistant surface areas. These strips are perpendicular to the taxiway or taxiway edge markings. In curved sections, the strips are placed on the tangency points of the curve and with a maximum interval of 15 m. On straight sections, the maximum interval is 30 m.

Closed taxiway marks:

They are yellow crosses of 90 cm width which indicate that a certain taxiway remains closed to aircraft traffic. At least one sign shall be placed at each end of the street or stretch that is intended to close.

Entry prohibited marks:

It is placed as a complement of the entry prohibited signs and located at the starting point of the prohibited area.

Information marks:

There are several basic types of information signs. They indicate the designation and direction of taxiways or other messages of interest to operations.

Mandatory marks:
They are presented on the pavement when it is not possible to install a sign and will contain the mandatory instructions to identify the place beyond which a taxiing aircraft or vehicle must not proceed unless authorised by the tower airport control. The background colour of the markings is reflective red.

3.2.7. Lights

Taxiway centreline lights:
Taxiway centreline lights are used at night or when the visibility is less than 350 m. They are fixed and green if the taxiway is not a runway exit but they will alternate green and yellow if so. These lights are separated 30 m on the straight sections.

Taxiway edge lights:
Blue taxiway edge lighting is also provided as additional guidance on taxiways as it is possible to operate at night. Each light point will be uniformly distributed with a 60 m separation between each other.

RETIL lights are not installed due to the absence of any rapid exit taxiway.

Stop bars lights:
Stop bars are located at the holding points and at the points where the traffic must stop and provide support to the marking. They are perpendicular to the taxiway, red coloured and uni-directional.

3.3. Apron

According to the characteristics of the airport, a paved apron has been conceived.

3.3.1. Slope

It is necessary to define a slope to prevent water accumulation.

On the one hand, taxiways on apron possess the same slope as the taxiways previously defined (section 3.2.2). On the other hand, parking stands will have a maximum slope of 1%.

3.3.2. Passenger terminal apron concept

Taking into consideration the operations managed by the airport, a remote apron has been designed.

There is no need to connect the terminal building with the aircraft via fingers. Instead, passenger transporters are used.

3.3.3. Service roads

The apron counts with service roads to enable ground personnel to move safely within it. A width of 10 m has been defined for them.

3.3.4. Parking stands

The airport reference code, as stated previously, defines a maximum wingspan of 36 m.

Subsequently, Type VIII parking stands, with a width of 37 m and a length of 34,5 m, are enough to accommodate the aircraft allowed in the airport.

Furthermore, aircraft perform autonomous manoeuvres to access and depart from the parking stands. Even though it requires providing more space for planes to manoeuvre, the traffic volume of the airport permits it and there is a saving in equipment and personnel.

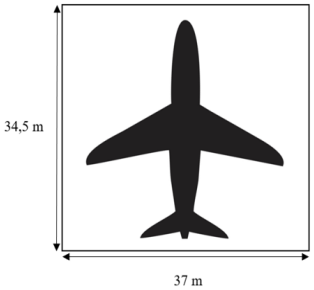


Figure 4. Type VIII parking stand.

3.3.5. Separation

The separations involving taxiways and taxilanes placed in the apron are already defined in section 3.2.3. Nevertheless, it is necessary to specify the distance which needs to be maintained between aircraft and other aircraft, buildings or objects, which has a value of at least 4,5 m.

3.3.6. Facilities

A de-icing/anti-icing facility is placed within the apron to service aircraft that need it.

Moreover, an isolated parking stand located more than 100 m away from other stands, buildings and public areas is conceived for emergency cases.

3.3.7. Horizontal markings

The apron is provided with the following markings:

Taxiway centreline:
Yellow line placed at the centre of the taxiways to guide aircraft while taxiing to the place where the aircraft stand markings are.

Apron edge:
Double yellow-coloured line which borders the apron surface capable of bearing the weight of the aircraft and serves as an extension of the taxiway edge sign.

Security line:
Red wide line which delimits the space designated for aircraft movement, separating it from areas intended for other purposes and which may contain obstacles for aircraft (such as parking stands or equipment storage).

Equipment restriction area:
Red line which delineates an equipment restriction area (ERA) or aircraft parking position within it. It is also known as ASA (aircraft safety area), denoting the space where the aircraft is parked and serviced. During the aircraft's entry manoeuvre, the ERA/ASA must be clear of vehicles, equipment, and personnel, who should maintain, generally, a minimum distance of 7,5 m from the aircraft.

Team waiting area:
Dashed white line which delineates an ESA equipment waiting area, which is a designated area situated at a safe distance from the parked aircraft, utilized by the handling teams to await the completion of the aircraft manoeuvre into the parking space.

Equipment parking area:
White line which delineates a zone for EPA equipment parking, accessible only via the dashed part of the line.

Parking prohibition area:
Framed diagonal red lines which delineate a strictly no-parking and no-stopping zone for NPA equipment.

Guide lines:
Parking stand entry lines:
Yellow-coloured line which, from the taxiway centreline, is used to guide the pilot to the parking position. It can be simple, straight or displaced.

Turn line:
Yellow-coloured line which assists the pilot in making turns into or out of the parking space.

Parking point departure line:
Yellow line which guides the pilot from parking stands to taxiways. It may have explicitly a turn or be simple, straight or displaced.

Reference bars:
Turn bar:
Yellow-coloured bar which indicates to the pilots the initiation of the turn. It is positioned to the left in the direction of travel, perpendicular to the guideline, with an arrow indicating the turn direction.

Nose wheel bar:
Yellow bar which shows to the personnel in the movement area where the nose gear wheels ought to be positioned when the aircraft is parked.

Stop bar:
Yellow bar which assists the pilot by indicating the stopping point. It is positioned to the left of the pilot.

Alignment bar:
Yellow-coloured bar which enables the pilot to properly align the aircraft at the conclusion of the parking manoeuvre.

Stand marking:
Directional mark to the aircraft stand:
Informative mark with yellow background which shows the direction towards the access of specific parking stands.

Access point to the stand:
Informative mark with yellow background which shows the point for accessing a parking stand.

Identification of the stand:
Informative mark with yellow letters and dark background which aids pilots in identifying their positions.

3.3.8. Lights

Aircraft stand manoeuvring guidance lights:
Fixed yellow lights located with the aircraft stand markings. Crucial for positioning aircraft in poor visibility conditions on paved aprons or de-icing/anti-icing facilities. The spacing between lights should not exceed 7,5 m on curves and 15 m on straight sections.

Stop position lights:
Fixed unidirectional red lights to indicate stop positions.

3.3.9. Ground equipment service

The most important servicing the apron should provide is pressure refuelling, external AC/DC electrical power, ground air conditioning, toilet servicing, housekeeping, potable-water and stairs and transport for passengers and baggage to move between the remote parking stand and the terminal building.

3.3.10. Taxiways and taxilanes

Taxiways, with a width specified in section 3.2.1, allow aircraft to move within the apron. Taxilanes, with the same width as taxiways, enable them to access and leave the parking stands.

3.3.11. Capacity and dimensions

When it comes to the dimensions of the apron, it is of paramount importance to analyse previously its capacity to design it efficiently.

Taking into consideration that all the parking stands are Type VIII, a single type of aircraft has been considered, the one corresponding to the reference aircraft.

To begin with, the average stay time of the reference aircraft in the parking stand ought to be taken into account: according to the technical characteristics of the plane, an average of 15 minutes are needed to perform the terminal operations [see the Appendix 5 for details], including the deplaning and enplaning of passengers and baggage, and the required maintenance.

To continue, the number of available parking stands must be determined. As the airport handles 28 arrivals and 28 departures per hour, 30 parking stands have been defined.

The following expression is used to compute the stand capacity:

$$F = \frac{P_i}{\Sigma\,(M_i \cdot T_i)} = \frac{30}{1 \cdot 0{,}25} = 120\ aircraft/h$$

Furthermore, the stay time fraction required by an aircraft of type i can be computed as follows:

$$t_i = \frac{M_i \cdot T_i}{\Sigma\,(M_i \cdot T_i)} = \frac{100 \cdot 0{,}25}{100 \cdot 0{,}25} = 1$$

To calculate the preliminary capacity of the stand, the expression presented below is used:

$$C = F \cdot X_{min} = F \cdot \left(\frac{p_i}{t_i}\right) = 120 \cdot 1 = 120\ aircraft/h$$

The percentage of arrivals is 50%. When it comes to the occupancy factor of the stand, a value of 80% has been considered.

To conclude, the capacity of the apron can be computed with the equation presented below:

$$C_a = \frac{F \cdot X_{min} \cdot U}{\%\ Arrivals} = \frac{120 \cdot 1 \cdot 80}{50} = 192\ aircraft/h$$

Where,

$F = stand\ capacity$

$P_i = number\ of\ positions\ of\ type\ i$

$M_i = \%\ of\ aircraft\ of\ type\ i\ over\ the\ total$

$T_i = stay\ time\ for\ an\ aircraft\ of\ type\ i$

$C = preliminary\ capacity$

$X_{min} = smallest\ value\ of\ p_i\ /\ t_i$

$p_i = fraction\ of\ positions\ of\ type\ i$

$t_i = stay\ time\ fraction\ for\ an\ aircraft\ of\ type\ i$

$U = occupancy\ factor$

As it can be observed, the capacity of the apron exceeds significantly the number of operations handled by the airport. Nevertheless, it has to be considered that flight classes are imparted and there are general aviation flights, so some stands will be used by planes dedicated to those purposes.

In addition, some parking stands are assigned to maintenance tasks or long stays. Therefore, it is usual having an apron capacity able to accommodate more planes than the ones operating commercial flights.

The 30 parking stands are placed linearly in two rows, maintaining the required clearance of 4,5 m.

The parking stands area has a length of 43,5 m to maintain the clearance in the front and rear of the plane. When it comes to the width, the wingspan of the reference aircraft is smaller enough to maintain the clearance between other aircraft parked next to it without the need of separating the parking stands. Consequently, the width of the rows of parking stands can be computed as follows:

$$Width = \frac{num.\ stands}{2} \cdot width\ stand\ type\ VIII = \frac{30}{2} \cdot 37 = 555\ m$$

Finally, two taxiways parallel to the runway connect the runway exits with the taxilanes, which surround the parking stands and enable the access and departure of planes from them.

4. Dimensioning of the landside

The landside is the area of an airport aimed to provide services to both passengers and planes. It consists of different buildings, each with a different purpose.

Regarding the depicted buildings' location, they are carefully distributed on the airport surface to avoid interfering with daily aircraft operations, [see the Appendix 6 for details].

4.1. Passenger terminal building

The passenger terminal building has the main purpose of transferring passengers and luggage between air transport to surface transport while accomplishing high-efficiency standards and requirements proposed by IATA and other recognized organisations.

The terminal building is directly connected to the national railway, bus station, and taxi services and also with parking for passengers and employees' vehicles to ensure a variety of connections between the airport and passengers' final destinations.

Because the chosen reference aircraft is only capable of allocating 56 passengers per flight, it has been decided that all commercial parking stands will be using remote parking stands. So internal bus services between the apron and the terminal building must be provided. [see the Appendix 7 for details regarding the interior of the terminal building].

4.2. Corporate terminal

The corporate terminal building has the aim of welcoming private business jets and is segregated from general and commercial aviation to ensure both privacy and exclusive access to the VIPs travelling from or to the airport.

4.3. Control tower

Due to the reduced size of the airport and to provide a detailed view of all the apron and runways, the control tower is located in the middle and at the top of the passenger terminal building.

4.4. Flight schools and general aviation

Because the services required by flight schools and general aviation are much different and simpler than the ones required by commercial and corporate aviation, a segregated area in the apron has been defined for allocating its much smaller planes.

4.5. Apron control access facility

To allow flight schools, general aviation authorised personnel and handling vehicles to directly access the apron, an access control facility has strategically been placed near its apron stand areas.

4.6. Emergency services

Due to the reduced airport dimensions, only one firefighting station must be considered to be able to certify response times anywhere in the airport.

4.7. Hangars

To allow airlines and private owners an interior area to park or perform maintenance works, some hangars have been built inside the airport area.

5. Definition of the landside areas and processes

To provide efficient passenger flux segregation, the terminal building is divided into three main areas: departures, services and arrivals.

The departures area has the purpose of checking in passengers and luggage, pursuing security checks, and passport controls and providing an area for passengers to wait for the bus to move them to their aircraft remote parking stand.

The service area comprehends the allocation of police dependencies, airport staff offices, and meeting rooms, connects both departure and arrival areas through a service corridor and also one for passengers taking connection flights, and the control tower on the upper floors of the building.

Regarding the arrivals area, it receives the buses with the arriving passengers, allows them to recover their checked luggage and also allocates the immigration controls before entering the country.

As ADRM by IATA establishes, the terminal building must supply enough space to accommodate all the necessary functions and processes comfortably while providing acceptable processing and waiting times for passengers, so the following design parameters have been considered for the following computations:

$$DPHP = 1568\, pax$$
$$PK = 50\%$$

5.1. Departure area processes

A flux diagram of the processes involved in the passenger departures area can be found in the appendices. [see the Appendix 8 for details].

5.1.1. Public departure hall

IATA establishes that for an optimum design level a minimum surface of 2 m^2 per passenger must be provided. The number of average passengers in the departing hall must be computed as:

$$P_D = \frac{DPHP \cdot T_p}{60} + \frac{DPHP \cdot VR_D \cdot T_V}{60}$$

Where T_p is the average time passengers spend at the hall, VR_D is the average visitor per departing passenger and T_v is the average time visitors spend at the hall. When assuming standard values measured at airports with similar characteristics, a T_p and a T_v of 5 minutes is obtained. Regarding the VR_D and because it is a regional airport, few visitors are expected, so 1 visitor per passengers has been taken as the average value to finally obtain:

$$P_D = \frac{1568 \cdot 5}{60} + \frac{1568 \cdot 1 \cdot 5}{60} = 261,33 \approx 262\ pax$$

$$Dept.\ Hall\ area = 2 \cdot P_D = 2 \cdot 262 = 524\ m^2$$

5.1.2. Self-service boarding pass/tagging kiosk

Self-service boarding pass kiosks allow passengers to directly check-in themselves by presenting a valid boarding pass, a passport or other similar documents.

The number of these kiosks can be computed as:

$$SS_{kiosks} = \frac{DPHP \cdot PK \cdot SS_{ratio} \cdot \frac{PT_{SS}}{60}}{30 + MQT} \cdot C_f$$

Where SS_{ratio} stands for the percentage of passengers using these terminals, PT_{SS} is the average processing time, MQT is the maximum queuing time and C_f is a correction factor for the queue.

As established by IATA, the introduced parameters are standard values taken from the ADRM. For this case and in order to maintain an optimum level of agreement the following values have been considered: SS_{ratio} equal to 10%, PT_{ss} 120 seconds, MQT equal to 3 minutes and C_f equal to 1,2.

$$SS_{kiosks} = \frac{1568 \cdot 0,5 \cdot 0,1 \cdot \frac{120}{60}}{30 + 3} \cdot 1,2 \approx 6\ kiosks$$

These kiosks will be distributed all among the departure halls and near the security filters. The area destined to these terminals can be obtained as:

$$A_{kiosks} = (SS_{kiosks} \cdot Kiosk_{area} \cdot AA) + (Q_{max} \cdot SP)$$

Where SS_{kiosks} stands for the number of kiosks, kiosk area equals 0,36, AA is equal to 3 and refers to the adjustment area for a single self-service kiosk and SP equals 1,5. Regarding Q_{max}, it can be easily computed as:

$$Q_{max} = Q_f \cdot DPHP \cdot PK \cdot SS_{ratio}$$

Where for this case Q_f is equal to 0,121 and SS_{ratio} is equal to 10%:

$$Q_{max} = 0,121 \cdot 1568 \cdot 0,5 \cdot 0,1 \approx 10\ pax$$

Finally, the total area for the kiosks can be computed as follows:

$$A_{kiosks} = (6 \cdot 0,36 \cdot 3) + (10 \cdot 1,5) \approx 22\ m^2$$

5.1.3. Check-in desk

The number of checking desks can be computed as:

$$CD = \frac{DPHP \cdot PK \cdot PT}{60} \cdot \frac{1}{30 + MQT} \cdot C_f$$

Where PT equals 140 seconds, MQT equals 20 minutes and C_f equals 1.

$$CD = \frac{1568 \cdot 0.5 \cdot 140}{60} \cdot \frac{1}{30 + 20} \approx 37\ checking\ desks$$

The area designated for the check-in desks can be obtained as:

$$A_{Check-in} = Q_{max} \cdot SP + CD \cdot A$$

$$Q_{max} = Q_f \cdot DPHP \cdot PK$$

By assuming SP equal to 1,4 m^2, the area of a checking desk itself to be 12 m^2 and Q_f equal 0,416:

$$Q_{max} = 0,416 \cdot 1568 \cdot 0,5 \approx 327\ pax$$

$$A_{Check-in} = 327 \cdot 1,4 + 37 \cdot 12 \approx 902\ m^2$$

5.1.4. Security checkpoint

For the security checkpoint and in order to guarantee appropriate waiting times, the number of security filters have been computed as:

$$SEC = \frac{DPHP \cdot PK \cdot PT}{60} \cdot \frac{1}{30 + MQT} \cdot C_f$$

Where PT equals 30 seconds, MQT equals 7 minutes and C_f equals 1,11:

$$SEC = \frac{1568 \cdot 0,5 \cdot 30}{60} \cdot \frac{1}{30 + 7} \cdot 1,11 \approx 12\ filters$$

The area designed for these purpose can be computed as:

$$A_{Security} = Q_{max} \cdot SP + SEC \cdot A$$

$$Q_{max} = Q_f \cdot DPHP \cdot PK$$

Where SP equals 1,2, Q_f equals 0,223 and the area of each security filters equals 20 m^2:

$$Q_{max} = 0,223 \cdot 1568 \cdot 0,5 \approx 175\ passengers$$

$$A_{Security} = 175 \cdot 1,2 + 12 \cdot 20 = 450\ m^2$$

5.1.5. Emigration (passport control)

For passport control, the number of desks have been computed as:

$$Emigration_{desks} = \frac{DPHP \cdot PK \cdot PT}{60} \cdot \frac{1}{30 + MQT} \cdot C_f$$

Where PT equals 30 seconds, MQT equals 7 minutes and C_f equals 1,11:

$$Emigration_{desks} = \frac{1568 \cdot 0{,}5 \cdot 30}{60} \cdot \frac{1}{30+7} \cdot 1,11 \approx 12\, desks$$

The area designed for these purpose can be computed as:

$$A_{Security} = Q_{max} \cdot SP + SEC \cdot A$$

$$Q_{max} = Q_f \cdot DPHP \cdot PK$$

Where SP equals 1,2, Q_f equals 0,223 and the area of each security filters equals 5 m^2:

$$Q_{max} = 0,223 \cdot 1568 \cdot 0,5 \approx 175\, pax$$

$$A_{emigration} = 175 \cdot 1,2 + 12 \cdot 5 = 270\, m^2$$

5.2. Arrivals area processes

A flux diagram of the processes involved in the passenger arrivals area can be found in the appendices. [see the Appendix 9 for details].

5.2.1. Immigration (passport control)

For passport control, the number of desks have been computed as:

$$Immigration_{desks} = \frac{DPHP \cdot PK \cdot PT}{60} \cdot \frac{1}{30 + MQT} \cdot C_f$$

Where PT equals 30 seconds, MQT equals 7 minutes and C_f equals 1,11:

$$Immigration_{desks} = \frac{1568 \cdot 0{,}5 \cdot 30}{60} \cdot \frac{1}{30+7} \cdot 1,11 \approx 12\, desks$$

The area designed for these purpose can be computed as:

$$A_{Security} = Q_{max} \cdot SP + SEC \cdot A$$

$$Q_{max} = Q_f \cdot DPHP \cdot PK$$

Where SP equals 1,2, Q_f equals 0,223 and the area of each security filters equals 5 m^2:

$$Q_{max} = 0,223 \cdot 1568 \cdot 0,5 \approx 175\, pax$$

$$A_{immigration} = 175 \cdot 1,2 + 12 \cdot 5 = 270\, m^2$$

5.2.2. Baggage claim area

Baggage claim area can be computed as the number of baggage carousels multiplied by the area occupied by one carousel.

$$Area_{baggage\ claim} = Carousels \cdot area_{carousel}$$

The area of one carousel has been considered to be equal to 30 m^2 and 3 carousels have been placed:

$$Area_{baggage\ claim} = Carousels \cdot area_{carousel}$$

$$Area_{baggage\ claim} = 5 \cdot 30 = 150\, m^2$$

5.2.3. Public arrivals hall

IATA establishes that for an optimum design level a minimum surface of 2 m^2 per passenger must be provided. Therefore, the number of average passengers in the arrivals hall must be computed as:

$$P_A = \frac{DPHP \cdot T_p}{60} + \frac{DPHP \cdot VR_D \cdot T_V}{60}$$

Where T_p is the average time passengers spend at the hall, VR_D is the average visitor per arriving passenger and T_v is the average time visitors spend at the hall. When assuming standard values measured at airports with similar characteristics, a T_p and a T_v of 5 minutes is obtained. Regarding the VR_D and because it is a regional airport, few visitors are expected, so 1 visitor per passengers has been taken as the average value to finally obtain:

$$P_A = \frac{1568 \cdot 5}{60} + \frac{1568 \cdot 1 \cdot 5}{60} = 261,33 \approx 262\, pax$$

$$Arrivals\ Hall\ area = 2 \cdot P_D = 2 \cdot 262 = 524\, m^2$$

5.3. Segregation of fluxes

As can be seen in the map of Appendix 10, the proposed design for the terminal provides a 100 % flux segregation between arriving and departing passengers.

References

[1] *ICAO Aerodrome Reference Code*. Available at: https://skybrary.aero/articles/icao-aerodrome-reference-code (Accessed: 16 April 2024).

[2] *BOMBARDIER Dash 8 Q300 (airport planning manual)*.

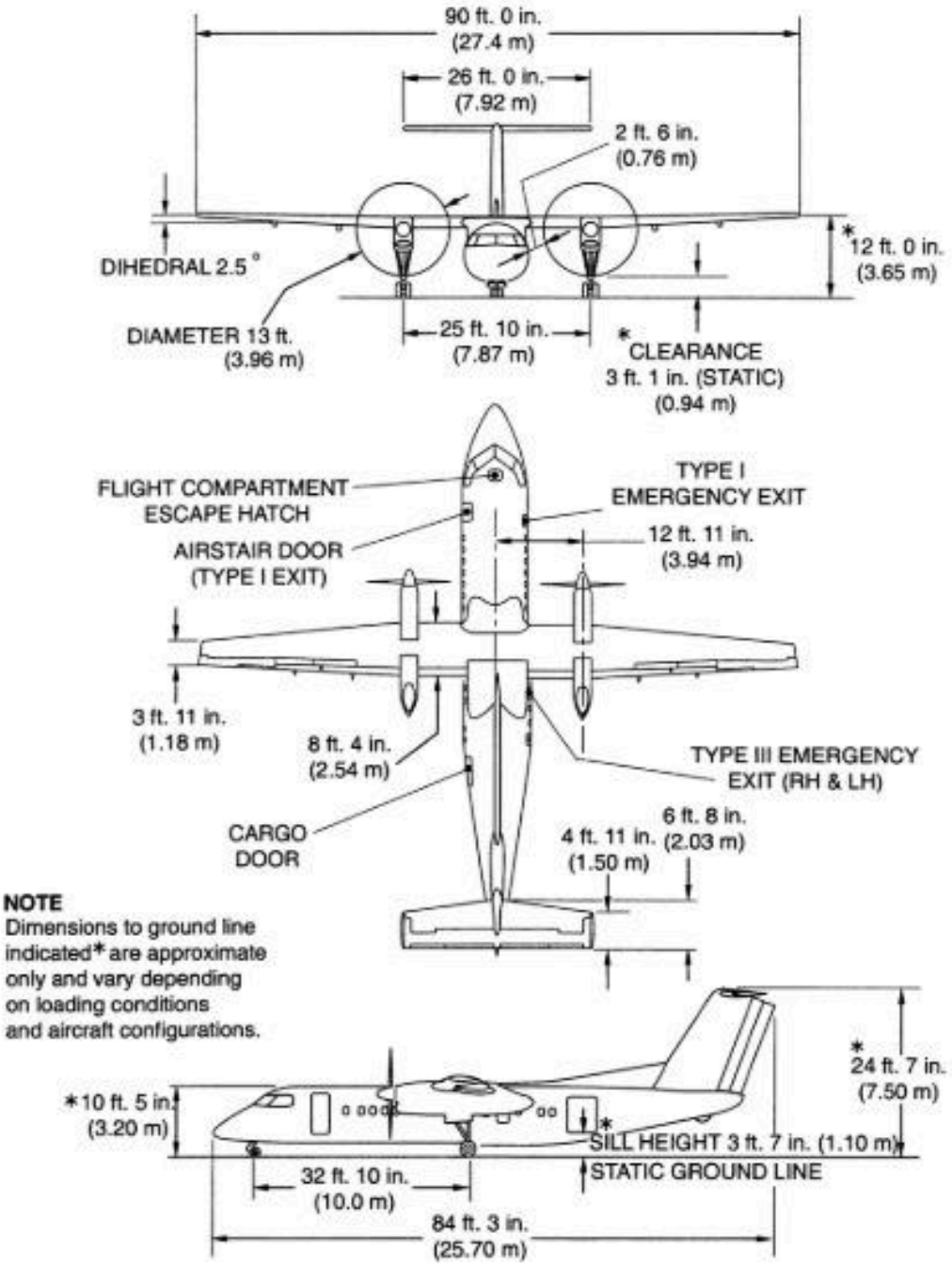
[3] *Aerodrome Design Manual (DOC 9157). Part I Runways*. Third Edition, 2006.

[4] *Inside Air New Zealand's World-Leading Dash 8-Q300 Operations*. Available at: https://simpleflying.com/air-new-zealand-worlds-largest-dash-8-300-airline/ (Accessed: 16 April 2024).

[5] *Manual Normativo de Señalización en Área de Movimiento*. Updated in 2001

Appendices

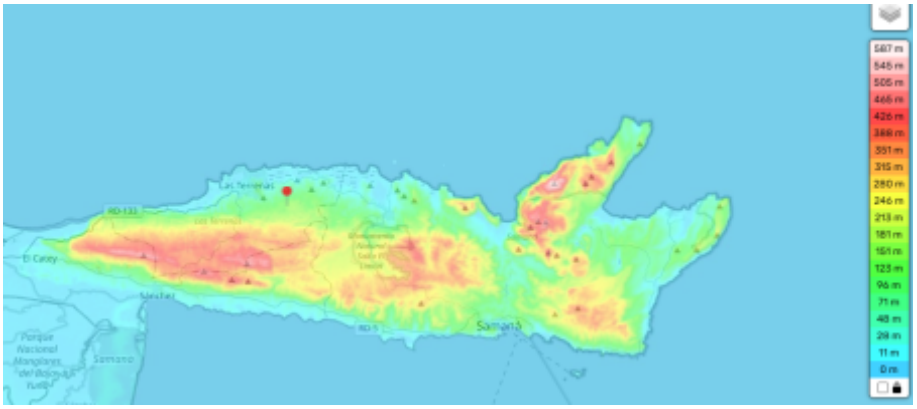
Appendix 1. Technical specifications of the reference aircraft:

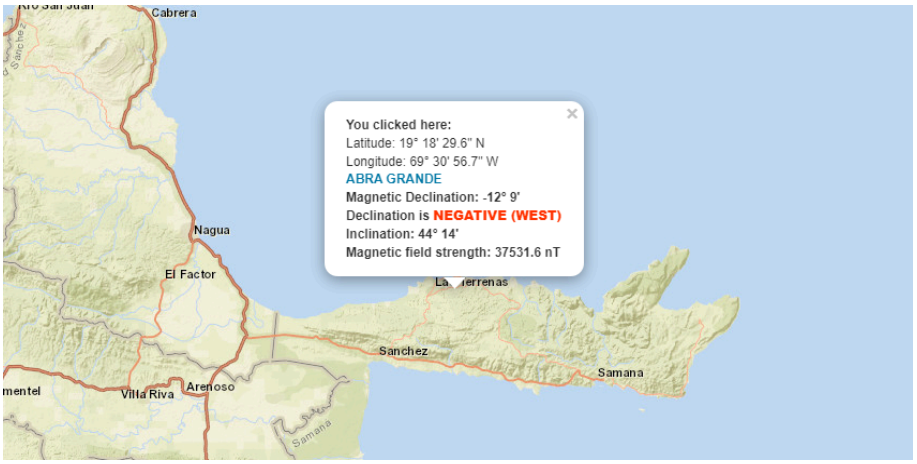


Appendix 2. Runway orientation:

Runway Orientation														
Primary Orientation		**Secondary Orientation**												
E	-1,57079633	**NNE**	-0,39269908											
								single RWY	**Primary Orientation**					**Cross RWY**
Speed kt	km/h		**Primary Orientation**				No use	**Utilization rate**	**Secondary Orientation**				No use	**Utilization rate**
>20	>37		0	0	0	3,25	3,25	96,75	0	0	0	0	0	100
>13	>24		0	1,75	8,25	6,25	16,25	83,75	0	0	3,5	2	5,5	94,5

Appendix 3. Terrain elevation:



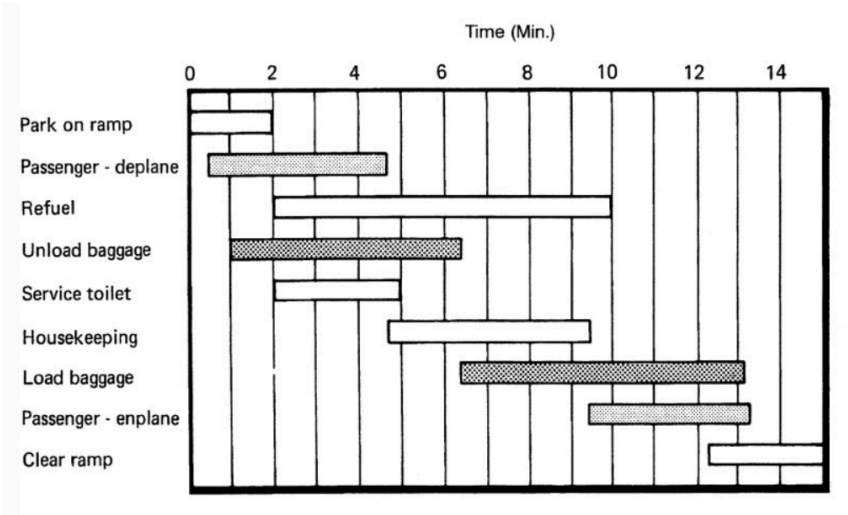


Appendix 4. Operations per hour for the designed airport:

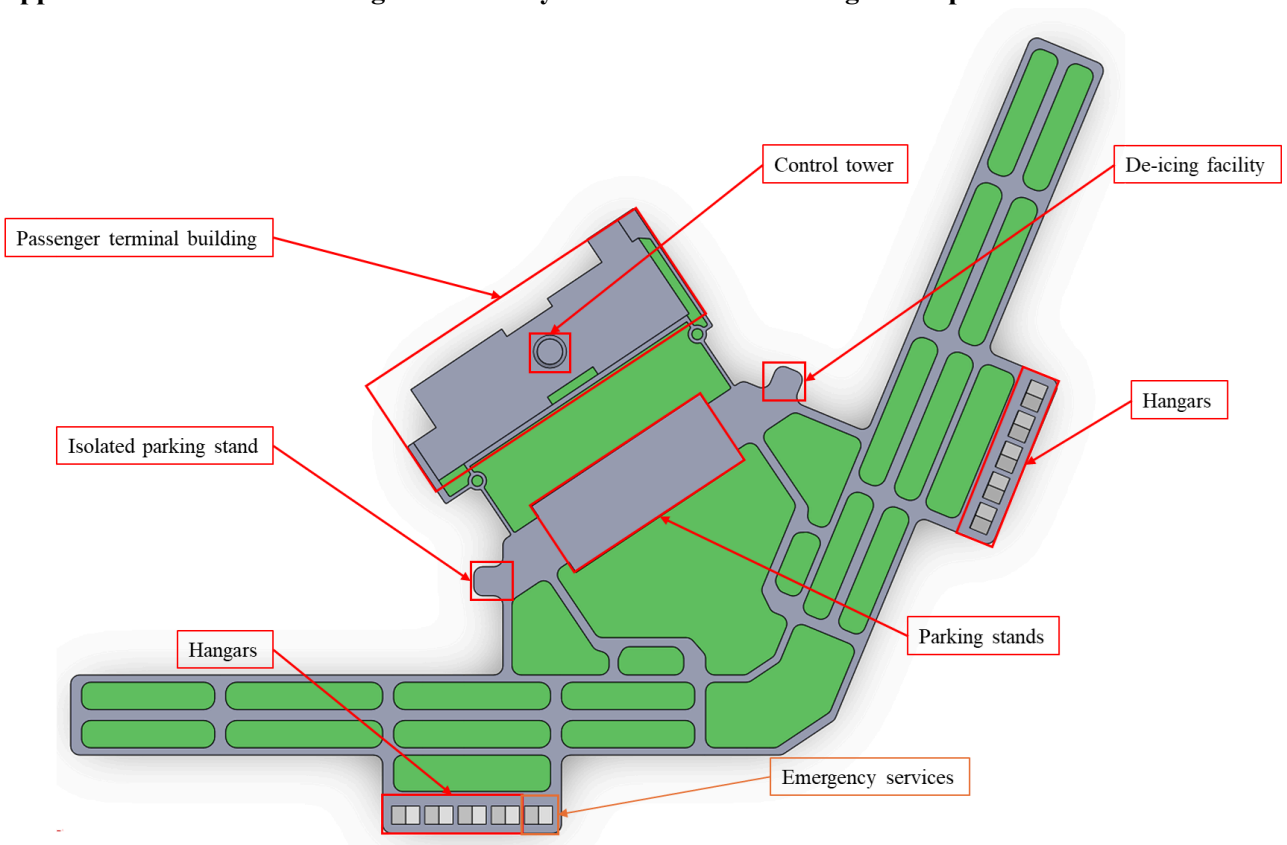
Operational efective time:	1,5 minutes

	Operations per hour with an open V runway configuration with full segregated operations							
	Arrivals				Departures			
	Commercial	Corporate	General Aviation	Flight Schools	Commercial	Corporate	General Aviation	Flight Schools
%	70%	5%	10%	15%	70%	5%	10%	15%
Number of operations	28,00	2,00	4,00	6,00	28,00	2,00	4,00	6,00

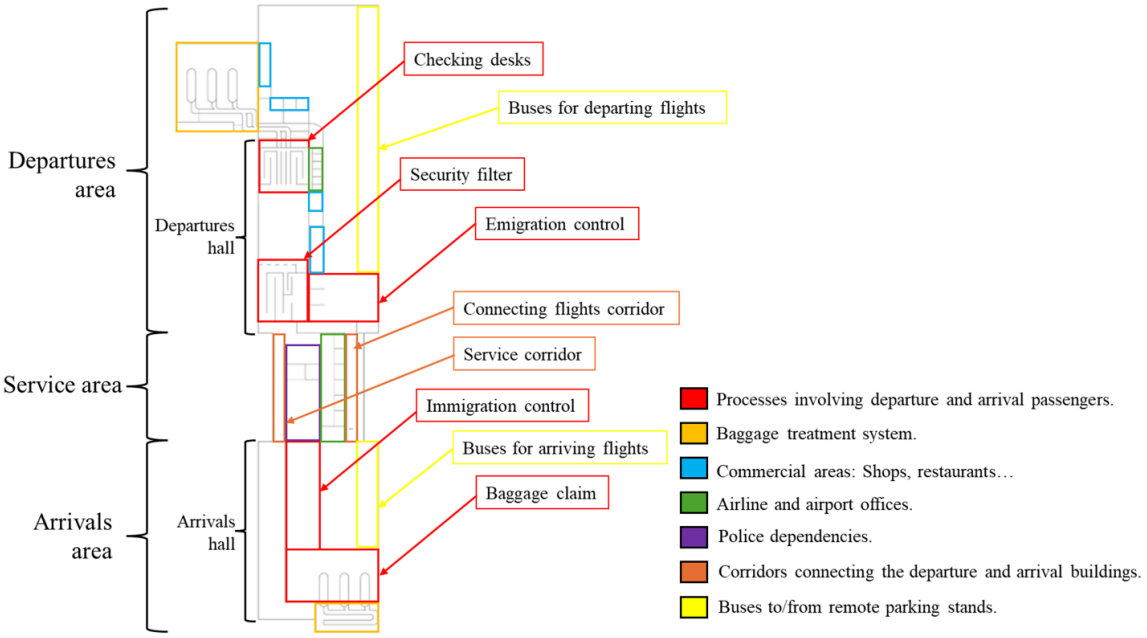
Appendix 5. Time distribution of the terminal operations required by the reference aircraft:



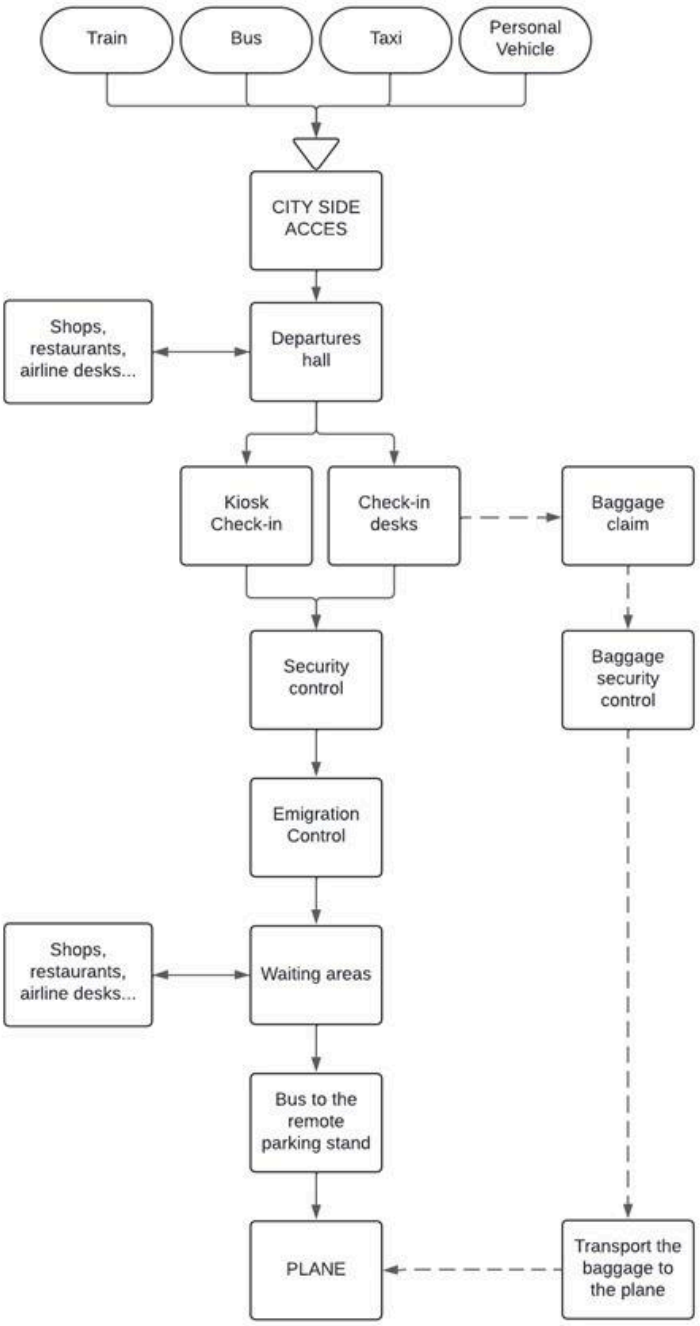
Appendix 6. Landside buildings and runway distribution of the designed airport



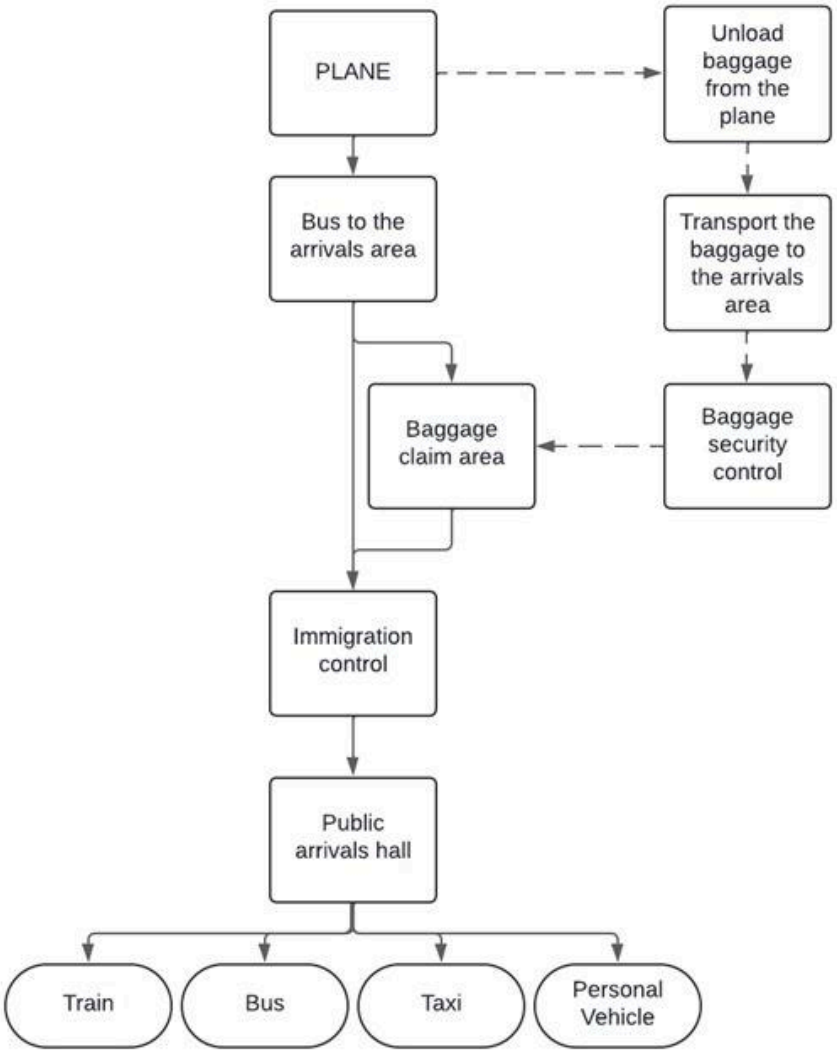
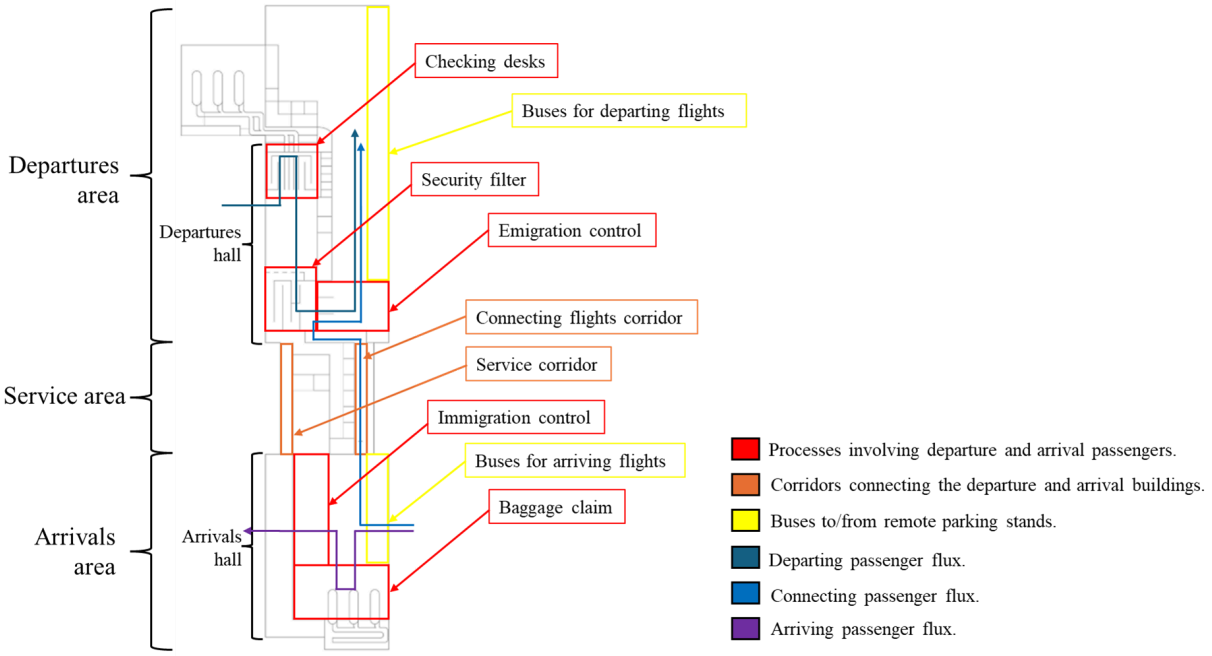
Appendix 7. Processes and areas location among the terminal



Appendix 8. Departure passengers flux and involved processes



Appendix 9. Arrival passengers flux and involved processes



Appendix 10. Segregation of fluxes map